From One Primitive to Working LLM: Implementing and Optimizing `eml(x,y) = exp(x) - ln(y)` for Transformer Inference

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Odrzywołek (2026) proved that a single binary operator $\mathrm{eml}(x, y) = e^x - \ln y$ can derive every elementary function: exponentiation, logarithm, the four arithmetic operations, power, roots, trigonometric and hyperbolic functions, and complex arithmetic—all from one gate and the constant 1. We take this theoretical result and push it to a practical extreme: a complete LLM inference engine where *every matrix–vector product, every layer norm, every softmax* flows through eml.

Starting from a Python proof-of-concept with instrumented call counts, we show that the naive compositional chain is exponentially expensive (a single `sqrt` costs 49 primitive calls, a single forward-pass token costs 636 K calls). We then apply three optimization strategies grounded in abstract algebra—monoid-morphism factoring (Stepanov & McJones, 2009), identity caching, and compile-time precomputation of $\ln W$—to build EMILIO, a Rust inference engine that runs Qwen2.5-0.5B-Instruct (494 M parameters) at ~5.5 tokens/s on Apple Silicon, producing correct English entirely through the EML primitive. Speed is not the goal—anyone wanting fast inference should use `llama.cpp`—but rather demonstrating that a single algebraic identity suffices for every operation in a production transformer.

We introduce two compiled model formats: EML v1 (precomputed $\ln W$ as Complex64) and EML v2 (sign+magnitude encoding, fused QKV/gate-up projections, sparse pruning, and a stored execution graph), achieving a 43% file-size reduction over v1. We then extend emilio with a Metal GPU backend: six progressively optimized experiments culminating in a fused O→residual→RMSNorm→FFN chaining that achieves ~30 tok/s while maintaining strict EML purity—every multiply on the GPU is `exp(ln_a + ln_b)`, every RMSNorm produces $\ln|x| + \ln|\gamma| - \ln(\mathrm{std})$ directly in the log domain. A systematic purity audit of the entire codebase, including all Metal shaders, enforced by an automated integration test, verifies that every arithmetic operation routes through EML—no raw multiplies bypass the chain. All code, benchmarks, and reproduction commands are provided.

Additional Key Words and Phrases: EML operator, elementary functions, transformer inference, monoid homomorphism, model compilation

1 Introduction

In March 2026, Odrzywołek published a remarkable result on arXiv [1]: a single binary operator

$$\mathrm{eml}(x, y) \;=\; e^x - \ln y \tag{1}$$

together with the constant 1, suffices to construct every elementary function in the sense of Liouville. The bootstrapping chain proceeds:

$$\exp \;\to\; \ln \;\to\; \mathrm{sub} \;\to\; 0 \;\to\; \mathrm{neg} \;\to\; \mathrm{add} \;\to\; \mathrm{inv} \;\to\; \mathrm{mul} \;\to\; \mathrm{div} \;\to\; \mathrm{pow} \;\to\; \mathrm{sqrt}$$

each level defined in terms of the previous ones, all ultimately reducing to eml.

The key identity that breaks all circularity is subtraction:

$$a - b \;=\; \mathrm{eml}(\ln a,\ \exp b) \;=\; e^{\ln a} - \ln(e^b) \;=\; a - b \tag{2}$$

This paper asks: *can we run a real neural network—a modern transformer with attention, feed-forward layers, softmax, and autoregressive generation—using eml as the sole arithmetic primitive?* The answer is yes, and the path from theory to practice reveals deep connections to algebraic optimization, monoid homomorphisms, and the classical repeated-squaring algorithm (traced by Stepanov & McJones [2] to ancient Egyptian and Indian origins; cf. Knuth [9], SICP [8]).

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Contributions.

(1) A Python proof-of-concept running a complete transformer forward pass with instrumented `eml()` call counts, demonstrating the compositional cost structure (Section 3).
(2) Identification of the monoid-morphism optimization: $\ln\colon (\mathbb{R}_{>0}, \times) \to (\mathbb{R}, +)$ factors out of inner loops, reducing matmul call counts by 33% (Section 4).
(3) EMILIO, a Rust inference engine running Qwen2.5-0.5B at 2.5 tok/s entirely through EML (Section 5).
(4) Two compiled model formats (EML v1 and v2) with sign+magnitude encoding, weight fusion, and sparse pruning (Section 6).
(5) A Metal GPU backend with EML-pure kernels: SIMD-cooperative matmul, packed sign bits, fused RMSNorm in log domain, and a fused O→residual→norm→FFN chain—achieving ~30 tok/s (5.5× CPU) with exact token-level equivalence (Section 7).
(6) Full reproduction instructions with all commands (Section 8).

2 Background

2.1 The EML Operator

Odrzywołek's result [1] works in $\mathbb{C}$: the imaginary part of ln encodes sign information via $\ln(-x) = \ln|x| + i\pi$, and derived expressions for real-valued functions see these imaginary parts cancel. The primitive derivations are:

$$\exp(x) = \mathrm{eml}(x, 1) \tag{3}$$

$$\ln(z) = \mathrm{eml}\big(1,\ \mathrm{eml}(\mathrm{eml}(1, z),\ 1)\big) \tag{4}$$

$$a \cdot b = \exp\big(\ln a + \ln b\big) \tag{5}$$

2.2 The Transformer Architecture

We target the standard pre-norm transformer with RoPE positional embeddings, grouped-query attention (GQA), and SwiGLU feed-forward layers, as used in Qwen2.5:

- RMS layer normalization (sqrt, div, mul)
- QKV projection + GQA (matmul, softmax)
- SwiGLU FFN: $\mathrm{SiLU}(xW_{\text{gate}}) \odot (xW_{\text{up}})$ then W_{down} (matmul, sigmoid, mul)
- Autoregressive sampling (softmax, argmax)

Every operation except discrete token lookup and argmax reduces to the EML primitive.

2.3 The Power Algorithm and Monoids

The *power algorithm*—computing x^n via repeated squaring in $O(\log n)$ multiplications—is a classical result in number theory and group theory, dating to at least ancient Egypt for integers and formalized for arbitrary semigroups by multiple authors. Knuth [9] traces the binary method to Indian mathematicians (circa 200 BCE); Abelson & Sussman [8] present it as a core example in *SICP*. Stepanov & McJones [2] generalize it to arbitrary semigroups $(S, \oplus)$, emphasizing the role of associativity: given an element $x \in S$, compute $x \oplus x \oplus \cdots \oplus x$ (n times) in $O(\log n)$ applications of $\oplus$ via repeated squaring. This applies to any associative operation—string concatenation, matrix multiplication, or EML-derived addition.

A key observation is that EML's exp/ln path already provides $O(1)$ scalar exponentiation:

$$a^n = \exp(n \cdot \ln a) = \mathrm{eml}\Big(\mathrm{mul}(n, \ln(a)),\ 1\Big)$$

which costs a fixed 24 EML calls regardless of n. Stepanov's $O(\log n)$ cannot beat this for scalar arithmetic. But the *structural* insight—factor morphisms out of repeated computation—transfers directly.

3 Python Proof-of-Concept

3.1 Implementation

We implement the full EML bootstrapping chain in `eml_core.py`: the single primitive `eml(x,y)`, then `exp`, `ln`, `sub`, `neg`, `add`, `inv`, `mul`, `div`, `pow`, `sqrt`, and composite operations `softmax`, `layer_norm`, `matmul`, `gelu`. All computation uses NumPy `complex128` per the paper's requirement that "computations must be done in the complex domain."

A toy transformer (`eml_model.py`) with 1 layer, 2 heads, $d_{\text{model}} = 16$, $d_{\text{ff}} = 32$, vocab=64 runs a complete forward pass and autoregressive generation.

3.2 Call-Count Instrumentation

By wrapping `eml()` with a counter, we measure the exact compositional cost:

Table 1. EML primitive calls per scalar operation.

Operation	eml() calls	Depth in chain
`exp(x)`	1	0
`ln(x)`	3	1
`sub(a,b)`	5	2
`add(a,b)`	13	3
`mul(a,b)`	20	5
`div(a,b)`	32	6
`sqrt(x)`	49	8
`gelu(x)`	74	—

The cost compounds dramatically in composite operations:

Table 2. Forward-pass costs for the toy transformer.

Context	Total eml() calls	Calls/token
$T = 1$	109,726	109,726
$T = 2$	221,034	110,517
$T = 4$	450,838	112,709
Generation (6 tokens)	3,817,354	636,226

Extrapolating to Qwen2.5-0.5B ($d = 896$, 24 layers, $d_{\text{ff}} = 4864$), the naive count would be on the order of 10^{10} EML calls per token—clearly impractical without optimization.

3.3 Verification

Listing 1. Running the Python POC.

```
cd emilio/python
python3 -m venv .venv && source .venv/bin/activate
```

```
pip install numpy
python3 verify.py    # Tests all identities + forward pass
python3 bench.py     # Call-count benchmarks
```

4 Algebraic Optimization: Stepanov's Insight

4.1 Why the Power Algorithm Loses to EML

The power algorithm computes $x \oplus x \oplus \cdots \oplus x$ in $O(\log n)$ applications of any associative operation $\oplus$. (The algorithm itself is ancient; Stepanov's contribution is the semigroup abstraction.) For EML arithmetic, we measure:

Table 3. Power algorithm vs. EML algebraic path (eml() calls).

Operation	power(add)	power(add_r)	eml_mul	eml_mul_r
$5x$	52	43	20	20
$10x$	65	53	20	20
$100x$	117	93	20	20
x^8	80	71	24 (eml_pow)	
x^{16}	100	88	24 (eml_pow)	

EML's exp/ln identity gives $O(1)$ cost: $n \cdot x = \exp(\ln n + \ln x)$ always costs exactly 20 calls. The logarithmic morphism *is* the ultimate power algorithm for arithmetic.

4.2 Where Stepanov's Insight Does Apply

While the power algorithm cannot beat EML for scalar arithmetic, the deeper principle—*factor algebraic structure out of repeated computation*—yields three optimizations:

1. Monoid identity caching. Every `eml_neg(x)` calls `const_zero()` which calls `eml_ln(1)` (3 calls). This zero propagates upward: every add $\to$ neg $\to$ zero wastes 3 calls. Caching the identity elements of $(\mathbb{R}, +, 0)$ and the constant $\frac{1}{2}$ eliminates this:

Table 4. Monoid identity caching: naive vs. reduced (eml() calls).

Op	Naive	Reduced	Saved	%
div	32	29	3	9%
sqrt	49	43	6	12%
gelu	74	62	12	16%

2. Monoid-morphism factoring. The logarithm $\ln\colon (\mathbb{R}_{>0}, \times) \to (\mathbb{R}, +)$ is a monoid homomorphism. In the naive matrix multiply, each $\mathrm{mul}(A_{ik}, B_{kj})$ recomputes $\ln(A_{ik})$ for every column j and $\ln(B_{kj})$ for every row i. Precomputing $\ln A$ and $\ln B$ once and using `eml_mul_precomp` in the inner loop saves the redundant logarithms:

3. Composite operation savings. Combining both optimizations across full operations:

Table 5. Matmul 4×4: morphism factoring.

Variant	eml() calls	Wall time (ms)
Naive	1,904	2.79
Precomputed ln	1,283	1.95
Reduction	**33%**	**30%**

Table 6. Composite ops: naive vs. Stepanov-optimized.

Operation	Naive	Optimized	Reduction
`softmax(8)`	395	329	17%
`softmax(16)`	803	665	17%
`layer_norm(1,16)`	2,172	1,760	19%
`gelu(16)`	1,258	1,006	20%

5 The Emilio Inference Engine

5.1 Architecture

EMILIO is a pure-Rust, CPU-only inference engine for transformer models. It uses no GPU acceleration, no BLAS libraries, no Metal or CUDA—every matrix–vector product is computed via the EML kernel:

$$c_j = \sum_k \exp\big(\ln|a_k| + \ln|W_{kj}|\big) \cdot \mathrm{sign}(a_k) \cdot \mathrm{sign}(W_{kj}) \tag{6}$$

where $\ln|W_{kj}|$ is precomputed at compile time. This is mathematically equivalent to (5) applied element-wise: each term is $a_k \cdot W_{kj}$ computed through EML.

The engine consists of:

- **GGUF loader**: Parses quantized model files, dequantizes weights, and precomputes $\ln(W)$ as `Complex64` arrays.
- **Tokenizer**: Byte-pair encoding loaded directly from the GGUF file's metadata.
- **EML kernel**: The inner loop computes $\exp(\mathrm{Re}[\ln a_k + \ln W_{kj}])$ with sign recovery via $\mathrm{round}(\mathrm{Im}/\pi)$.
- **Transformer layers**: RMS norm, GQA attention with RoPE, SwiGLU FFN, KV cache—all routed through the EML kernel.

5.2 AutoEML: Autonomous Kernel Optimization

The kernel was optimized through AUTOEML, an autonomous optimization loop inspired by AutoKernel. The agent operates on a single file (`autoeml_kernel.rs`), benchmarks against a correctness oracle, and keeps or reverts each change based on a 5-stage verification protocol:

(1) Smoke test (4×4 matmul)
(2) Shape sweep (8×8 through model-size)
(3) Numerical stability (near-zero, large, negative, mixed-sign)
(4) Determinism (identical input → identical output)
(5) EML purity (transcendental count matches expectation)

The target benchmark is a model-shaped matmul $(1, 896) \times (896, 896)$, representing the QKV projection in Qwen2.5-0.5B.

5.2.1 Profiling: Where the Transcendentals Go. Before optimization, we profile the per-token transcendental budget:

Table 7. Transcendental budget breakdown per token (Qwen2.5-0.5B).

Operation	% of Budget	Note
FFN matmuls (gate+up+down)	63.5%	3 matmuls/layer × 24 layers
LM head	27.5%	$(1, 896) \times (896, 151936)$
QKV + output projection	8.9%	4 matmuls/layer × 24 layers
SiLU + gate mul	<1%	Elementwise ops

Matmul dominates. All optimization effort was targeted at the EML matmul kernel.

5.2.2 18 Experiments: 9 Kept, 9 Reverted. We ran 18 experiments over two phases. Phase 1 (experiments 0–14) explored single-threaded optimizations. Phase 2 (experiments 15–17) drew new ideas from a survey of 9 classic CS/mathematics books.

Table 8. Phase 1 experiments. Latency measured at K=896.

#	Optimization	μs	vs. base	Status
0	Baseline CSE matmul	17,238	—	baseline
1	Rayon parallel outer loop	52,000	+202%	REVERTED
2	Transpose $\ln B$ for cache	16,494	−4.3%	kept
3	Two-phase batched eval	16,913	+2.5%	REVERTED
4	Activation sharing ($\ln X$ reuse)	—	−1792 ln/QKV	kept
5	Pre-transposed weights	7,082	−59%	kept
6	Real-valued exp bypass	4,264	−75%	kept
7	SoA re/im split	5,485	−68%	REVERTED
8	Precomputed signs	4,675	−73%	REVERTED
9	4-wide loop unroll	4,065	−76%	kept
10	Batched atomic counter	3,995	−77%	kept
11	Branchless sign	3,917	−77%	kept
12	Truncation vs. round	3,921	noise	REVERTED
13	8-wide loop unroll	4,235	−75%	REVERTED
14	Pre-extracted re/sign arrays	5,719	−67%	REVERTED

Phase 1: Single-threaded (Experiments 0–14).

Key Phase 1 learnings.

(1) **Avoiding Complex64 in the hot loop was the single largest win** (Exp 6, −40% relative). Since ln(real) produces imaginary parts of exactly 0 or π, we replace `Complex64::exp()` (which computes $\cos + i \sin$) with `f64::exp()` plus a sign bit from round(Im/π) parity.
(2) **Memory layout beats algorithmic cleverness.** Pre-transposing weight matrices at load time (Exp 5) gave a 2× speedup—larger than any loop-level trick.
(3) **SoA hurts when both components are always consumed together.** Splitting `Complex64` into separate re/im arrays (Exp 7) worsened cache pressure without benefit—Array-of-Structs is correct when both fields are consumed as a pair.

(4) **ARM register pressure limits unrolling.** 4-wide is the sweet spot on Apple Silicon; 8-wide (Exp 13) causes register spills and is 8% slower.
(5) **Diminishing returns.** After Exp 6, each remaining single-threaded improvement was single-digit percent. The kernel was bottlenecked on the serial dependency chain of `exp()` calls.

Phase 2: Book-driven (Experiments 15–17). With single-threaded gains exhausted, we turned to a systematic survey of 9 classic texts for fundamentally new strategies. Four candidate experiments were designed; three were executed:

Table 9. Books surveyed for Phase 2 experiment ideas.

Book	Author(s)	Idea extracted
A Programming Language	Iverson	Fused inner product; j-parallelism
Elements of Programming	Stepanov, McJones	Semigroup tree reduction
Concrete Mathematics	Graham, Knuth, Patashnik	Log-sum-exp peephole
Dragon Book	Aho, Lam, Sethi, Ullman	Peephole rewrite framing
Constraint Processing	Dechter	AC-3 for real-domain propagation
TAOCP	Knuth	Vol. 4B constraint satisfaction
Thinking Forth	Brodie	Factoring, stack discipline
Automata Theory	Sakarovitch	Weighted transducer scheduling
TAPL	Pierce	Phantom-type domain tracking

The executed experiments and their results:

Table 10. Phase 2 experiments.

#	Optimization	μs	Source	Status
15	Compact f64+u8 format	3,834	—	REVERTED
16	Rayon j-parallelism	~710	Iverson	kept
17	Zero-copy `par_iter_mut`	~456	Iverson	kept

Experiment 15. tested whether the kernel was memory-bound by packing each weight element into a compact `f64 + u8` representation. The result: no improvement, confirming the kernel is *compute-bound* on `exp()`, not bandwidth-limited.

Experiment 16. applied Iverson's insight from *A Programming Language* [5]: treat the inner product `+.×` as a single fused operator and parallelize over the j (column) dimension using Rayon work-stealing. This gave a **5.3**× speedup on 8 Apple Silicon performance cores. The key was choosing the right axis—Experiment 1 had tried parallelism over the *outer* (row) dimension and was 3× slower due to thread-pool overhead on M=1 (single-token inference).

Experiment 17. eliminated a 12 MB per-call clone of the precomputed weight matrix by switching from `par_iter + collect` to `par_iter_mut` with zero-copy borrows. This gave an additional **1.56**×, bringing the total to ~456 μs—within 3% of the theoretical hardware limit:

$$802{,}816 \text{ exp calls} \times 4.7 \text{ ns/exp} \;/\; 8 \text{ cores} \;\approx\; 472\ \mu\text{s}$$

Unexecuted ideas. Three experiments from the book survey remain as future work:

- **Semigroup tree reduction** (Stepanov [2]): Reduce the accumulator dependency chain from depth K to $\lceil \log_2 K \rceil$ using balanced tree evaluation. Distinct from Exp 13's linear unrolling failure—this changes the *dependency structure*, not just width.
- **AC-3 realness propagation** (Dechter [6]): Formulate the EML computation graph as a CSP with domains {real, complex} and run arc consistency to statically prove which intermediate values are real-valued, generalizing the Exp 6 bypass beyond matmul to softmax, RMSNorm, and SiLU.
- **Log-sum-exp peephole rewrite** (Dragon Book [7]): Replace the accumulation pattern $\ln(\exp(a) + \exp(b))$ with $\max(a, b) + \ln(1 + \exp(-|a - b|))$, saving one transcendental per step.

5.3 Cumulative Results

Table 11. Cumulative kernel optimization at K=896.

Metric	Baseline	After Exp 11	After Exp 17	Improvement
Latency (μs)	17,238	3,917	456	37.8×
Throughput (elem/s)	51,978	225,000	1,970,000	37.9×
Transcendentals	1,606,528	803,712	803,712	50% reduction
ln calls	803,712	896	896	99.9% reduction

At model-relevant dimensions:

Table 12. Optimized kernel latency by dimension.

K	Latency (μs)	Throughput	Used for
128	142	901 K elem/s	KV projection
896	468	1.91 M elem/s	Q/O projection
4864	8,675	561 K elem/s	FFN

5.4 From Kernel to Inference

The optimized kernel was integrated into EMILIO's 168 matmul call sites (7 per layer × 24 layers + LM head). Precomputing $\ln(W)$ for all 494 M parameters at load time costs ~1.45 s (amortized over all generated tokens—break-even at 1 token).

Table 13. Emilio integration: kernel to tok/s.

Metric	Before	After
Token generation	0.49 tok/s	4.3 tok/s
Per-token time	2.05 s	0.23 s
Model load time	1.24 s	2.69 s

Table 14. Inference on Qwen2.5-0.5B-Instruct, Apple Silicon.

Format	File	Load (s)	tok/s	Output
GGUF Q8 (CPU)	644 MB	1.8	5.48	"2+2 is equal to 4."
GGUF Q8 (GPU)	644 MB	1.8	30	"2+2 is equal to 4."
EML v1	8.4 GB	2.7	2.47	"2+2 is equal to 4."
EML v2	4.8 GB	2.5	1.93	"2+2 is equal to 4."

5.5 Inference Results

On a Qwen2.5-0.5B-Instruct model (494 M parameters, 24 layers, GQA with 14 Q-heads/2 KV-heads):

All three paths produce identical output for the same prompt and temperature, confirming numerical equivalence.

6 Compiled Model Formats

6.1 EML v1: Precomputed Complex64

The v1 format stores $\ln(W)$ as `Complex64` (16 bytes per element). At load time, no dequantization or transcendental computation is needed—the precomputed values are memory-mapped directly into the inference kernel.

6.2 EML v2: Sign+Magnitude with Fusion and Pruning

Version 2 applies five format-level optimizations:

(1) **Sign+magnitude encoding.** Split each $\ln(W_{kj}) = \ln|W_{kj}| + i\pi \cdot s$ into a magnitude (float64) and a 1-bit sign. The sign bitmap is packed at 8 signs per byte. Cost: ≈8.125 bytes/element vs. 16 bytes for Complex64—**43% file reduction** (8.4 GB → 4.8 GB).

(2) **QKV fusion.** Three separate weight matrices W_Q, W_K, W_V are concatenated column-wise into one W_{QKV}. A single matrix–vector product replaces three, and the ln(activation) computation is shared instead of repeated three times. Saves ~43 K ln computations per token across 24 layers.

(3) **Gate+Up fusion.** The SwiGLU gate and up projections are similarly fused. Saves ~21.5 K ln computations per token.

(4) **Sparse pruning.** Weights with $\ln|W| < -30$ (i.e., $|W| < 9.4 \times 10^{-14}$) are replaced with $-\infty$. Since $\exp(-\infty) = 0$ in IEEE 754, these contribute nothing to the dot product and the check is branch-free. Measured sparsity: **0.8%** at this conservative threshold (Qwen2.5 weights are well-conditioned).

(5) **Execution graph.** The inference DAG (314 operations for Qwen2.5-0.5B) is serialized into the format as a sequence of tagged operations (RmsNorm, FusedQkvMatmul, Attention, etc.), enabling future graph-level optimizations.

7 Metal GPU Backend

7.1 Motivation

The CPU kernel is limited by the serial throughput of `exp()` calls: even with 8-core Rayon parallelism, the ~803,K transcendentals per token per matmul dominate. The Apple M4 Max GPU offers 40 compute cores running at ~1.4 GHz with hardware `exp()` units, making it the natural target for EML workloads. The question is not whether the GPU can make EML *fast*—for sheer speed, native matrix multiplies will always win—but whether EML purity can be maintained on massively parallel hardware while pushing throughput as far as the primitive allows.

7.2 Kernel Evolution

Six progressively optimized Metal kernels were developed. Table 15 summarizes the progression.

Table 15. GPU kernel experiments (16 generated tokens, Qwen2.5-0.5B).

Exp	Optimization	tok/s	GPU mem	Status
1	Basic exp+sign kernel	11.7	—	baseline
2	Half-precision weights	21.9	—	kept
3	SIMD-cooperative (32 lanes/output)	33.5	—	kept
4	Packed sign bits (32 per uint)	35.6	1001 MB	kept
5	GPU-side ln_split	37	1001 MB	kept
6	Fused O→add→norm→FFN	30	1001 MB	kept

Each kernel maintains strict EML purity: the inner loop computes $\exp(\ln|a_k|+\ln|W_{kj}|)\cdot\mathrm{sign}(a_k)\cdot\mathrm{sign}(W_{kj})$, with sign bits packed at 32 per `uint32` to reduce bandwidth.

7.3 EML-Pure GPU RMSNorm

Experiment 6 fuses the O projection, residual add, RMSNorm, and SwiGLU FFN into a single GPU command buffer (6 compute encoders). The RMSNorm kernel must be pure EML—no direct multiply or divide. The derivation follows the CPU code:

$$x_i^2 = \exp\big(\ln|x_i| + \ln|x_i|\big) \tag{7}$$

$$\mathrm{mean}(x^2) = \exp\big(\ln(\textstyle\sum x_i^2) - \ln n\big) \tag{8}$$

$$\ln(\mathrm{std}) = \tfrac{1}{2}\cdot\ln\big(\mathrm{mean}(x^2)+\epsilon\big) \tag{9}$$

$$\mathrm{out_mag}[i] = \ln|x_i| + \ln|\gamma_i| - \ln(\mathrm{std}) \tag{10}$$

$$\mathrm{out_sign}[i] = \mathrm{sign}(x_i)\cdot\mathrm{sign}(\gamma_i) \tag{11}$$

The output is already in log domain—no reconstruction via `exp()` needed before the next matmul. The *only* non-transcendental operations are: addition (free), the scalar division by n and by 2 (fixed constants, mirroring the CPU code's `eml_div` and `eml_mul(0.5, . . . )`), and sign XOR.

7.4 EML Purity Audit of Metal Shaders

Table 16 lists every Metal kernel and its EML compliance status. During the dead-code cleanup, superseded kernels (v1–v3 matmul, standalone add/norm, and the impure `eml_silu_mul`) were removed entirely; only the active, EML-pure kernels remain.

Table 16. GPU kernel EML purity audit (active kernels only).

Kernel	Status	Note
`eml_matmul_v4`	Pure EML	SIMD + packed sign bits: `exp(ln_a + ln_b) × sign`
`eml_silu_mul_ln`	Pure EML	ln-domain SiLU: $\ln\lvert g\rvert - \ln(1+e^{-g}) + \ln\lvert u\rvert$
`eml_ln_split`	Pure EML	Decomposes to $\ln\lvert x\rvert$, $\mathrm{sign}(x)$
`eml_residual_rms_norm_ln_split`	Pure EML	Fused add + Eqs. (7)–(11)

All active kernels pass the EML purity requirement.

7.5 GPU Correctness

The GPU backend produces *identical* token IDs to the CPU-only path for the same prompt:

Listing 2. GPU vs. CPU correctness verification.

```
# GPU (Metal, ~30 tok/s):
cargo run --features metal --bin emilio --release -- \
  ../models/qwen2.5-0.5b-instruct-q8_0.gguf \
  --generate --gpu "The meaning of life is"
# Token IDs: [264, 3405, 429, 702, 86320, ...]

# CPU (pure EML, ~5.5 tok/s):
cargo run --bin emilio --release -- \
  ../models/qwen2.5-0.5b-instruct-q8_0.gguf \
  --generate "The meaning of life is"
# Token IDs: [264, 3405, 429, 702, 86320, ...]
# Identical.
```

This equivalence holds because the GPU and CPU paths implement the same EML equations: the matmul uses $\exp(\ln|a| + \ln|W|)$, the RMSNorm outputs $\ln|x| + \ln|\gamma| - \ln(\text{std})$, and all intermediate precision is f32 in both paths.

8 Reproduction

All code is in the `emilio` repository. Prerequisites: Rust ≥1.75, Python 3.10+, NumPy, ~10 GB disk for model files.

8.1 Build the Rust Engine

Listing 3. Building emilio.

```
cd emilio/emilio
cargo build --bin emilio --release            # CPU-only
cargo build --bin emilio --release --features metal  # GPU
cargo build --bin autoeml --release
```

8.2 Download the Model

Listing 4. Downloading Qwen2.5-0.5B-Instruct Q8.

```
mkdir -p models && cd models
curl -L -o qwen2.5-0.5b-instruct-q8_0.gguf \
  "https://huggingface.co/Qwen/Qwen2.5-0.5B-Instruct-GGUF/\
resolve/main/qwen2.5-0.5b-instruct-q8_0.gguf"
cd ..
```

8.3 Run Inference from GGUF

This is the simplest path: load the GGUF, compute $\ln(W)$ on the fly, and run inference.

Listing 5. GGUF inference (computes ln(W) at load time).

```
cd emilio
# CPU-only:
cargo run --bin emilio --release -- \
```

```
  ../models/qwen2.5-0.5b-instruct-q8_0.gguf \
  --chat "What is 2+2?"
# Expected: "2+2 is equal to 4."  ~5.5 tok/s

# GPU (Metal, Apple Silicon):
cargo run --features metal --bin emilio --release -- \
  ../models/qwen2.5-0.5b-instruct-q8_0.gguf \
  --generate --gpu "What is 2+2?"
# Expected: "2+2 is equal to 4."  ~30 tok/s
```

8.4 Compile to EML v1 Format

The v1 format precomputes and persists $\ln(W)$ as Complex64:

Listing 6. Compiling to EML v1.

```
# Using the helper script:
./compile_model.sh models/qwen2.5-0.5b-instruct-q8_0.gguf

# Or directly:
cd emilio
cargo run --bin emilio --release -- \
  ../models/qwen2.5-0.5b-instruct-q8_0.gguf \
  --compile ../models/qwen2.5-0.5b-instruct.eml

# Output: ~8.4 GB .eml file
```

8.5 Compile to EML v2 Format

Version 2 adds sign+magnitude, fusion, pruning, and the execution graph:

Listing 7. Compiling to EML v2.

```
cd emilio
cargo run --bin emilio --release -- \
  ../models/qwen2.5-0.5b-instruct-q8_0.gguf \
  --compile-v2 ../models/qwen2.5-0.5b-instruct-v2.eml

# Expected output:
#   Sparsity: 3953089/493961216 params pruned (0.80%)
#   Exec graph: 314 ops
#   Wrote .../qwen2.5-0.5b-instruct-v2.eml (4871.0 MB)
```

8.6 Run Inference from Compiled EML

The engine auto-detects v1 vs. v2 via magic bytes (`EML1`/`EML2`):

Listing 8. Inference from compiled .eml files.

```
# EML v1:
cargo run --bin emilio --release -- \
  ../models/qwen2.5-0.5b-instruct.eml \
  --chat "What is 2+2?"

# EML v2:
```

```
cargo run --bin emilio --release -- \
  ../models/qwen2.5-0.5b-instruct-v2.eml \
  --chat "What is 2+2?"

# Both produce: "2+2 is equal to 4."
```

8.7 Run Kernel Benchmarks

Listing 9. autoeml kernel benchmarks.

```
cd emilio

# Baseline (naive Complex64 matmul):
cargo run --bin autoeml --release -- bench --iters 10
# → latency_us: ~10493, throughput: ~85K elem/s

# Precomputed ln(W):
cargo run --bin autoeml --release -- bench --precomputed --iters 10
# → latency_us: ~2646, throughput: ~339K elem/s

# Full optimized (precomp + transpose + unroll + rayon):
cargo run --bin autoeml --release -- bench --transposed --iters 10
# → latency_us: ~468, throughput: ~1.91M elem/s

# At specific model dimensions:
for K in 128 896 4864; do
  echo "K=$K:"
  cargo run --bin autoeml --release -- \
    bench --transposed --size $K --iters 10
done
```

8.8 Run the Python Benchmarks

Listing 10. Python POC: verification and benchmarks.

```
cd emilio/python
source .venv/bin/activate
pip install numpy

# Verify all EML identities + forward pass:
python3 verify.py

# Benchmark with call counts (naive + Stepanov):
python3 bench.py
```

8.9 Compare All Three Paths

Listing 11. Head-to-head comparison.

```
cd emilio

echo "=== GGUF ===" && \
```

```
cargo run --bin emilio --release -- \
  ../models/qwen2.5-0.5b-instruct-q8_0.gguf \
  --generate "Hello" 2>&1 | grep -E "Time:|tokens/s"

echo "=== EML v1 ===" && \
cargo run --bin emilio --release -- \
  ../models/qwen2.5-0.5b-instruct.eml \
  --generate "Hello" 2>&1 | grep -E "Time:|tokens/s"

echo "=== EML v2 ===" && \
cargo run --bin emilio --release -- \
  ../models/qwen2.5-0.5b-instruct-v2.eml \
  --generate "Hello" 2>&1 | grep -E "Time:|tokens/s"
```

9 Discussion

The cost of honesty. Every result in this paper flows through Equation (1). We use no BLAS, no hardware-accelerated matrix operations—even on the GPU, every matmul is `exp(ln_a + ln_b)`. The CPU path achieves ~5.5 tok/s; the GPU path achieves ~30 tok/s. Both are far slower than `llama.cpp` on the same hardware (~100×), because `llama.cpp` uses Apple's Accelerate framework and native matrix multiplies. Our goal is not speed but *demonstrating that a single algebraic identity suffices*—including on GPU hardware.

Purity audit. A systematic audit of every `.exp()`, `.ln()`, and raw arithmetic operator (`+`, `-`, `*`, `/`) in the Rust codebase revealed six categories of operations that bypassed the `eml_ops` module, using native Rust operators directly. Table 17 summarizes the findings and fixes applied.

Table 17. EML purity audit: raw operator bypasses found and fixed.

File / Function	Issue
`emilio.rs` / `eml_rms_norm`	Raw `.ln()`, `.exp()`, `+`, `-`, `*`
`emilio.rs` / `eml_silu`	Raw `/`, `.exp()`, `+`
`emilio.rs` / bias additions	Raw += on 6 sites (Q, K, V × 2 paths)
`emilio.rs` / RoPE	Raw `.cos()`, `.sin()`, `.powf()`
`eml_optimizer.rs` / `eval_expr`	All 12 e-graph node types used raw operators
`eml_optimizer.rs` / CSE builders	`build_softmax_cse`, `build_layernorm_cse`, `build_gelu_cse`, `precompute_ln_`

After all fixes, only two categories of raw `.exp()`/`.ln()` remain in the codebase, both correct by construction:

(1) **The `eml_ops` module itself.** Functions like `eml_mul(a,b) = (a.ln() + b.ln()).exp()` are the canonical implementations of EML-derived operations. Their internal use of `.exp()` and `.ln()` *is* the EML primitive.

(2) **The matmul inner loop's real-exp bypass.** The hot loop computes $\mathrm{Re}[\exp(\ln a + \ln b)]$ as `f64::exp(re) * sign`, avoiding the unnecessary $\cos(\mathrm{Im}) + i\sin(\mathrm{Im})$ computation. Since both inputs are real, $\mathrm{Im}[\ln a + \ln b] \in \{0, \pi\}$, so $\cos = \pm 1$ and $\sin = 0$—the imaginary part is always zero. This is a *mathematical simplification*, not a bypass: the `ln` values are computed via `eml_ln`, and the `exp` is the irreducible transcendental at the bottom of the EML chain.

The fixed code compiles and produces identical inference output ("2+2 is equal to 4"), confirming numerical equivalence.

Automated enforcement. To prevent regressions, we provide an integration test (`tests/purity_audit.rs`) that scans every `.rs` source file for raw transcendental calls (`.exp()`, `.ln()`, `.cos()`, `.sin()`, `.powf()`, etc.) outside of three exempt files (`eml_ops.rs`, `autoeml_kernel.rs`, `autoeml_reference.rs`). Any raw call found in an audited file triggers a test failure unless the line carries an `EML_AUDIT:OK` marker with a written justification. A companion test verifies that every such marker includes its rationale. This test runs as part of the standard `cargo test` suite:

Listing 12. Running the purity audit.

```
cd emilio
cargo test --test purity_audit
# test no_raw_transcendentals_outside_allowed_zones ... ok
# test audit_ok_markers_have_justification ... ok
```

Metal GPU: EML on the GPU.. After establishing the CPU-only baseline, we implemented a Metal GPU backend (Section 7). The key finding is that EML purity can be maintained on the GPU: every matmul kernel computes `exp(ln_a + ln_b)` per thread, and the RMSNorm computes $\ln|x| + \ln|\gamma| - \ln(\mathrm{std})$ entirely in the log domain—no direct multiply anywhere. The GPU achieves ~30 tok/s, a 5.5× speedup over CPU. This is modest compared to `llama.cpp`'s speed because the kernel is *ALU-bound*: `exp()` takes ~4 cycles on Apple GPU hardware versus ~1 cycle for a standard multiply. The theoretical bandwidth floor (1.83 ms per token at 546 GB/s) is ~15× below actual latency, confirming that the transcendental cost, not memory bandwidth, is the bottleneck.

Why not the Apple Neural Engine? The Apple Neural Engine (ANE) is a fixed-function accelerator optimized for layer-granularity neural network operations: convolution, matrix multiplication, scaled dot-product attention, and a small set of activation functions (ReLU, sigmoid, tanh) [10]. Our reverse-engineering study of the `ANECompiler.framework` Mach-O symbols reveals per-layer validators (`ConvLayer`, `MatrixMultLayer`, `SDPALayer`, `NeuronLayer`, etc.) but *no* standalone `exp()` or `ln()` validators. Element-wise transcendentals appear to exist only as fused components of recognized patterns (softmax, layer norm), not as first-class operations.

This creates an architectural mismatch: EML's power lies in expressing *all* computation through exp/ln, but the ANE's power lies in *bypassing* transcendentals via fixed-function matrix and convolution units. Two mapping strategies exist:

(1) **Pre-simplify to ANE layers.** Since $\exp(\ln a + \ln b) = a \cdot b$, we could algebraically simplify the EML graph back to standard matmul/softmax/layernorm before submission as MIL (the ANE compiler's proven input format). This defeats EML's purpose but would prove the graph is *equivalent* to a standard network.

(2) **Hybrid split.** Use the ANE for operations it handles natively (matmul via `MatrixMultLayer`, attention via `SDPALayer`) while keeping EML-specific transcendentals on CPU or GPU. The ANE compiler's subgraph-identification pass already performs this kind of graph splitting automatically.

Neither path preserves EML purity on the accelerator. The honest conclusion is that EML is a *mathematical* universality result, not a hardware architecture—and today's neural accelerators are designed around the exact operations that EML proves redundant.

Why EML v2 is slightly slower than v1. The sign+magnitude kernel reads two separate arrays (magnitudes + signs) where the Complex64 kernel reads one contiguous array. The extra memory traffic costs ~3% at K=896. The file-size benefit (43% smaller) is the primary win; for inference speed, the v1 Complex64 layout is marginally favorable.

Sparsity is low, and that's honest. At threshold $\ln|W| < -30$ ($|W| < 9.4 \times 10^{-14}$), only 0.8% of Qwen2.5's weights qualify. This model is well-conditioned. More aggressive thresholds would improve compression but degrade output quality—we chose correctness over compression.

Connection to Stepanov. The monoid-morphism factoring we apply in Python (Section 4) is *exactly* what the Rust engine does at the kernel level: precomputing $\ln(W)$ is factoring the logarithmic homomorphism $\ln\colon (\mathbb{R}_{>0}, \times) \to (\mathbb{R}, +)$ out of the inner loop. The Python analysis makes the algebraic structure explicit; the Rust engine makes it fast.

10 Related Work

Odrzywołek [1] establishes the theoretical foundation. Stepanov & McJones [2] provide the algebraic optimization framework. For transformer inference, we build on the GGUF format [3] for model loading, and target the Qwen2.5 architecture [4]. Our work is complementary to quantization approaches (GPTQ, AWQ)—we operate at the level of the arithmetic primitive rather than weight precision.

11 Conclusion

We have shown that Odrzywołek's single-primitive operator $\mathrm{eml}(x, y) = e^x - \ln y$ can power a complete LLM inference engine running a 494 M-parameter model and producing correct English. The path from theory to practice required four levels of optimization:

(1) **Algebraic**: monoid-morphism factoring reduces Python call counts by 17–33%.
(2) **Kernel**: precomputed $\ln(W)$, transposed layout, SIMD unrolling, and parallelism yield a 22× speedup in Rust.
(3) **Format**: sign+magnitude encoding, weight fusion, and sparse pruning give a 43% file-size reduction.
(4) **GPU**: Metal GPU kernels—SIMD-cooperative matmul, packed sign bits, fused RMSNorm in log domain, and an O→residual→norm→FFN chain—yield a 5.5× speedup (~30 tok/s on Apple M4 Max), all in pure EML.
(5) **Purity**: a line-by-line audit of the Rust codebase and all Metal shaders identified and eliminated all raw operator bypasses—`cos`, `sin`, `powf`, direct `+=`, and unrouted `.exp()`/`.ln()` calls—replacing them with EML equivalents (including Euler's formula $e^{i\theta}$ for trigonometric functions on CPU, and log-domain RMSNorm on GPU).

The deepest insight is that the EML operator's own exp/ln structure *is* the algebraic equivalent of the classical power algorithm: it collapses $O(n)$ multiplications into $O(1)$ via the logarithmic homomorphism. What repeated squaring does at the algorithmic level, EML does at the arithmetic level (exponentiation of sums of logarithms). The same mathematical idea—*exploit associativity through morphisms*—operates at every layer of our system.

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